

Vasant Kumar Desai

Clinical Data Programmer & Data Engineer | Python · R · SQL · CDISC SDTM · GxP
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PROFESSIONAL SUMMARY

Clinical Data Engineering professional with 7+ years of experience in clinical research and regulated environments (GxP, ALCOA+, ICH-GCP). Specializes in Python and R programming for clinical data cleaning, transformation, and review workflows aligned with CDISC SDTM/ADaM standards. Proven track record of building audit-ready, traceable data pipelines for clinical trials, with deep expertise in SQL, data quality checks, automated reporting, and cross-functional collaboration with Data Managers, Statisticians, and Medical Reviewers. Experienced with EDC systems, database lock deliverables, and contributing to inspection-ready environments.

CORE COMPETENCIES

Clinical Programming: Python (Pandas, NumPy), R (tidyverse, data.table, R Markdown), SQL

Clinical Data Standards: CDISC SDTM/ADaM concepts, CRF data processing, Data Management Plans (DMP)

Regulatory & Compliance: GxP, ALCOA+, ICH-GCP, 21 CFR Part 11, Audit Readiness, Validated Programming

Data Quality & Review: Data cleaning, reconciliation, query resolution, trend analysis, QC checks

Automation & Reporting: Reusable utilities, automated reporting pipelines, KPI dashboards, R Markdown

Platforms & Tools: Snowflake, dbt, Databricks, AWS (Glue, S3, Athena, SageMaker), Power BI

Other: Git, REST APIs, EDC/CDMS workflows, Apache Iceberg

WORK EXPERIENCE

Data Engineer III | Technical Lead | Thermo Fisher Scientific

Bengaluru, India | Nov 2025 – Present

- Develop enterprise data APIs and microservices providing secure, governed access to scientific and clinical datasets, applying ALCOA+ traceability principles.
- Contributing to large-scale migration of research and operational data to AWS data lake with automated validation, reconciliation, and audit logging.
- Building GenAI-powered applications for document summarization and semantic search over research data, enabling faster data review cycles.

Senior Data Engineer | Technical Lead | TEKsystems Global Solutions

Bengaluru, India | Oct 2024 – Nov 2025

- Led a 5-member data engineering team, defining technical roadmap and mentoring engineers on production-grade pipeline best practices.
- Architected Medallion Architecture (Bronze → Silver → Gold) on S3 + Apache Iceberg, enabling reproducible, analysis-ready datasets with full schema lineage.
- Developed internal data APIs to expose curated, validated datasets to ML and analytics teams — analogous to clinical data delivery workflows.
- Standardized ingestion workflows with schema registry, lineage tracking, RBAC access controls, and audit logging aligned with enterprise governance standards.
- Built SLA monitoring dashboards and automated alerts, reducing pipeline downtime by 40% and ensuring reliable data delivery to downstream consumers.
- Optimized Spark jobs (memory tuning, partitioning, compression), improving pipeline runtime by 30% and cutting infrastructure costs.

Data Delivery & Automation Specialist | Labcorp Laboratories India

Bengaluru, India | Apr 2021 – Sep 2024

- Built scalable Python and SQL pipelines for clinical trial data ingestion, transformation, and governed sharing across study teams — directly supporting Data Management Plan (DMP) deliverables.
- Performed clinical data quality checks, trend analysis, and issue identification to ensure accuracy and audit readiness; supported ongoing query resolution workflows.

- Developed governed clinical data warehouse with schema enforcement, versioning, RBAC, and validation workflows fully aligned with GxP and ALCOA+ principles.
- Engineered automated document intelligence platform (AWS Lambda + Textract) processing 20K+ clinical documents/month, improving compliance traceability.
- Designed predictive modelling pipelines in AWS SageMaker to forecast clinical trial milestones and resource bottlenecks.
- Automated KPI reporting dashboards for clinical teams using Python and Power BI, reducing decision-making time by 25%.
- Collaborated closely with clinical operations, regulatory, and compliance teams to produce SOPs, validation plans, and functional specifications for audit-ready deliverables.
- Built monitoring and data quality checks ensuring reproducibility and reliability of clinical datasets across interim and final database lock activities.

Junior Research Fellow | COE RVCE

Bengaluru, India | Aug 2018 – Mar 2021

- Developed and automated large-scale bioinformatics analysis workflows using Python, R, and Bash for reproducible research in regulated academic environments.
- Built reproducible genomic data pipelines and integrated public biological databases (Ensembl, dbSNP) for downstream statistical analysis.
- Open-sourced a PyMOL plugin adopted globally for protein analysis and visualization; published peer-reviewed research in clinical/biological data domains.

EDUCATION

M.Tech in Bioinformatics — R.V. College of Engineering, Bengaluru | Jun 2018

B.E in Biotechnology — M.S. Ramaiah Institute of Technology, Bengaluru | Jun 2015

CERTIFICATIONS

- AWS Certified Data Engineer – Associate (2024)
- AWS Certified Machine Learning Engineer – Associate (2025)
- Google Business Intelligence Specialization – Coursera
- Data Governance with Databricks – Coursera
- AutoML and BERT NLP – Coursera / LinkedIn

AWARDS & ACHIEVEMENTS

- Spot Award – Jun 2025 | TEKsystems Global Solutions
- Ace Award Level 4 – Automation & Process Optimization | Labcorp Laboratories India
- Ace Award Level 3 – Vendor Analytics Tool Development | Labcorp Laboratories India
- Outstanding Performance Rating – 2024 | Labcorp Laboratories India

PUBLICATIONS

- Desai, V. – Novelty of Data Mining Techniques for Bioinformatics Approach. Journal of Environmental Research and Development, Jun 2017.
- Desai, V. – Identification of Banana Heat Responsive Long Non-Coding RNAs and Their Expression Analysis. 3Biotech Journal.